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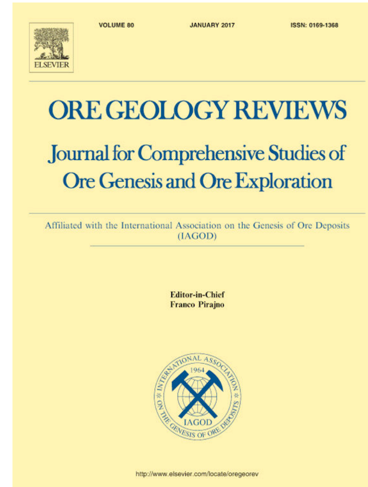

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Various occurrences and differential enrichment of Ge in sphalerite: An example from the Shanshulin Pb-Zn deposit, SW China

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Abstract

Germanium (Ge) is commonly enriched in low-temperature Pb-Zn deposits. However, the distribution and enrichment mechanisms of Ge vary significantly for individual deposits. This study employs SEM and LA-ICP-MS analysis techniques to investigate the occurrence and substitution of Ge within sphalerite, as well as to identify the factors controlling Ge enrichment. In the Shanshulin Pb-Zn deposit, three generations of sphalerite were identified: early stage (Sp1a + Sp1b) and late stage (Sp2). Sp1a is dark brown, and Sp1b is light brown, often in close association with or enclosing pyrite. Sp2 is light yellow to white and typically found in close contact with galena. LA-ICP-MS time-resolved depth-intensity profiles reveal that Ge can exist either within the sphalerite lattice or as Ge-rich nanoparticles. The substitution mechanisms for Ge incorporation into the sphalerite may include $Ge^{4+}+2(Ag^{+}, Cu^{+}) \leftrightarrow 3Zn^{2+}$ and $Ge^{2+} \leftrightarrow Zn^{2+}$ or $Ge^{4+} + \square$ (□ represents vacancy) $\leftrightarrow 2Zn^{2+}$. In the Shanshulin deposit, the chemical reactions and mass exchanges between gaseous sulfur species and minerals or fluids may induce variations in the temperature and fS_2 of ore-forming fluids. The concentrations of Fe, Mn, Cu, and Ga in sphalerite are influenced by temperature, while the concentration of Ge is jointly regulated by the temperature and composition of the ore-forming fluids. The difference in Ge enrichment among different Pb-Zn deposits in the northwestern Guizhou is controlled by temperature which influences the substitution mechanisms of Ge.

Key words: Germanium, Sphalerite, Substitution, Enrichment, Shanshulin Pb-Zn deposit

1. Introduction

Germanium (Ge) plays a crucial role in numerous high-tech fields, including semiconductors, infrared optics, and fiber-optic communications, making it an irreplaceable strategic resource in emerging industries. It has been classified as a critical metal (Höll et al., 2007; Fougerouse et al., 2023). About 75% of the world's Ge comes from Pb-Zn deposits (Meng et al., 2024), where it can be extracted as a by-product. With the rapid growth of science and technology, the global demand for Ge is constantly increasing. To ensure the sustainable development of Ge resources, it is imperative to understand the distribution, occurrence, and enrichment mechanisms of Ge in sphalerite (Liu et al., 2023; Sun et al., 2024). Previous studies have shown that Ge can enter sphalerite through isomorphism (Cook et al., 2009; Belissont et al., 2014; Meng et al., 2024). Various substitution mechanisms have been proposed based on correlations between trace elements, such as $3Zn^{2+}\leftrightarrow Ge^{4+}+2Cu^{+}$ (Wei et al., 2019), $4Zn^{2+}\leftrightarrow 2Fe^{2+}+Ge^{4+}+$ vacancy (Yuan et al., 2018), and $2(Zn^{2+}, Cd^{2+}) \leftrightarrow Mn^{2+}+Ge^{2+}$ (Hu et al., 2021). The diversity of substitution mechanisms reflects the complexity of interactions between elements in sphalerite. In addition, Ge-rich nanoparticles and independent Ge minerals have also been identified in some Pb-Zn deposits (Fougerouse et al., 2023; Meng et al., 2023; Sun et al., 2024), suggesting that the occurrence forms of Ge in Pb-Zn deposits are also diverse and require further research. Moreover, Ge shows differential enrichment across different sphalerite stages and within individual sphalerite particles (Belissont et al., 2014; Wei et al., 2019; Luo et al., 2021), the reasons for this differential enrichment are still not well understood. In summary, although it is well-recognized that Ge tends to be enriched in sphalerite, a comprehensive understanding of its occurrence, substitution, and differential enrichment in sphalerite is still lacking.

In the Sichuan-Yunnan-Guizhou (SYG) Pb-Zn metallogenic province, carbonate-hosted Pb-Zn deposits are widely developed. The primary ore-bearing strata are the Late Ediacaran Dengying Formation and Early Carboniferous strata (Liu and Lin, 1999). Pb-Zn ore bodies are fault-controlled, primarily occurring in layered and veined forms with high Pb (5 wt.%) and Zn (10 wt.%) grades, associated with critical elements such as Ge, Ga, In and Cd. The ore-forming temperatures range from 150°C to 270°C, and the ore-forming fluids are basin brines (Han et al., 2007; Tan et al., 2019; Xiong et al., 2019). This makes the region a natural laboratory and a hot spot for studying the metallogenic theory and mineralization of Ge. Using the Shanshulin carbonate Pb-Zn deposit in the SYG Pb-Zn metallogenic province as a case study, this research conducted a detailed mineralogical investigation using optical microscopy and Scanning Electron Microscope (SEM). Trace element concentrations in different generations of sphalerite were determined by Laser Ablation-Inductively Coupled Plasma-Mass Spectrometry (LA-ICP-MS). Subsequently, this study explored the occurrence, substitution mechanisms of Ge in the Shanshulin Pb-Zn deposit, as well as the controlling factors of element enrichment. Moreover, a model for the enrichment of trace elements in sphalerite was established.

2. Regional geology

The SYG Pb-Zn metallogenic province is located on the southwest margin of the Yangtze block (Fig. 1A), situated at the intersection of the Tethys and circum-Pacific metallogenic domains. The exposed strata within the Yangtze block include Precambrian basement and an overlying sedimentary cover. The basement consists of deeply metamorphosed Early Proterozoic-Archean crystalline rocks and a Meso-Neoproterozoic folded basement (Yuan et al., 1986). The lithology of this basement primarily comprises plagioclase gneiss, amphibolite, biotite granulite, migmatized gneiss, and medium to low-grade metamorphic marine volcanic and sedimentary rocks (Liu and Lin, 1999; Huang et al., 2004; Gao et al., 2008). The sedimentary cover is in unconformable contact with the basement, consisting mainly of shallow marine carbonate rocks and clastic sedimentary rocks (Yan et al., 2003). The Ediacaran-Ordovician, Carboniferous, Permian, and Triassic strata are thick and widely distributed. The strata from the Late Triassic to the Cenozoic were formed entirely in a continental sedimentary environment (Yan et al., 2003; Xiong et al., 2023). During the Middle-Late Permian, intense and frequent volcanic activities in this region led to the formation of the renowned Emeishan Large Igneous Province, covering an area of more than 250,000 km^2 (Zhou et al., 2002; Jian et al., 2009). Additionally, a series of tectonic events and numerous deposits were formed under the influence of the Indosinian movement (Yan et al., 2003; Zhang et al., 2006). These deposits were later affected and modified by the Yanshanian and Himalayan orogenic movements (Liu and Lin, 1999; Zaw et al., 2007). The NW-trending Ziyun-Yadu fault, SN-trending Anninghe-Lvzhijiang fault, and NE-trending Mile-Shizong fault are the primary fault structures in the SYG Pb-Zn metallogenic province (Fig. 1B). These faults extend to the basement and exhibit multi-stage activity influenced by tectonic movements during the Hercynian, Indosinian, and Yanshanian periods (Wei et al., 2019). These primary fault structures, along with numerous secondary structures, control the distribution of deposits, strata thickness, and paleogeographic environment in the three Pb-Zn ore concentration areas of northwestern Guizhou, northeastern Yunnan, and southwestern Sichuan (Liu and Lin, 1999; Han et al., 2007; Zhou et al., 2013). For example, the Shanshulin, Qingshan, Shaojiwan, Yadu and Tianqiao Pb-Zn deposits are closely associated with the NW-trending Yadu-Mangdong fault and the NW-trending Weining-Shuicheng fault, both of which are part of the Ziyun-Yadu fault system (Fig. 1B).

3. Geology of the Shanshulin deposit

The Shanshulin Pb-Zn deposit is situated in the southeast section of the Weining-Shuicheng fault zone (Zheng, 1992). It is one of the key deposits in northwest Guizhou

(Yang et al., 2018). In the Shanshulin deposit, the exposed stratigraphic sequences are mainly Carboniferous, Permian, and Lower-Middle Triassic strata (Fig. 2A). The Carboniferous strata predominantly comprise limestone, dolomitic limestone, and dolostone, along with a minor quantity of claystone. The Permian strata are mainly constituted by shale, coal, and limestone. Basalt is sandwiched between the Lower Permian and Upper Permian strata. The Lower to Middle Triassic strata consist mainly of sandstone, shale, mudstone, and marlstone. The ore bodies are mainly hosted in the Upper Carboniferous Huanglong Formation (Chen et al., 1984; Fu et al., 2004).

There are faults and folds in the Shanshulin deposit area (Jin, 2008). The F_1 fault is composed of several steeply dipping reverse faults with angles ranging from 50° to 80° (Zhou et al., 2014). The NW-trending (F_3) and NNW-trending (F_5) faults are normal faults (Zhou et al., 2014). The Guanyinshan syncline and anticline structures are located to the north and south of the F_1 fault (Fig. 2A). A total of twenty-two ore bodies have been identified in the Shanshulin deposit area (Zheng, 1992; Zhou et al., 2014). The ore bodies mainly concentrate within the fault zone between F_{11} and F_{30} (Fig. 2B). These ore bodies can be categorized into two types: layered and veined (Fig. 3 A, B). Veined ore bodies typically occur in the upper or lower parts of layered ore bodies (Zhou et al., 2014). The ore volume of the veined ore bodies is less than 0.2 Mt (Zhou et al., 2014). Among them, the grade of Pb ranges from 0.13 to 3.45 wt.% (average 2.16 wt.%), and the grade of Zn ranges from 0.85 to 5.65 wt.% (average 3.52 wt.%). The largest layered ore body No. 4, which accounts for 85% of the total metal reserves, extends 460 meters in length, reaches a depth of 145 meters, and varies in thickness from 0.19 to 17.79 meters (Zhou et al., 2014). The ore bodies, which contain at least 2 Mt of sulfide ores, have an average grade of 3.64wt % for Pb and 14.98wt % for Zn (Zhou et al., 2014).

The Shanshulin deposit features three types of ore: primary sulfide ores, secondary oxidized ores, and sulfide-oxidized mixed ores (Zhou et al., 2014). The primary sulfide ore is the dominant type, with smaller amounts of oxidized ore and sulfide-oxidized mixed ore found near the surface. The mineral composition of the sulfide ore is relatively simple, with the primary metallic minerals being sphalerite, pyrite, and galena, and gangue minerals including calcite, dolomite, and minor quartz. In contrast, the oxidized ores and sulfide-oxidized mixed ores have more complex mineral assemblages, including sphalerite, pyrite, galena, siderite, cerussite, and limonite (Zhou et al., 2014). Hydrothermal alteration in the deposit is relatively simple, mainly consisting of early dolomitization and pyritization, and late-stage calcitization (Zhou et al., 2013).

4. Samples and methods

4.1 Samples

Ore samples were collected from mining tunnels at elevations of 1300 m and 1720 m within the Shanshulin deposit. The samples were cleaned, cut, and polished to facilitate microscopic examination of mineral associations and ore textures. Prior to LA-ICP-MS analysis, the samples were observed using a SEM to detect any mineral inclusions or pores on the surface, minimizing the impact of contamination on the distribution of trace elements.

4.2 SEM analysis

Back-scattered electron (BSE) images of the ore samples were obtained at the State Key Laboratory of Ore Deposit Geochemistry (SKLODG), Institute of Geochemistry, Chinese Academy of Sciences (IGCAS). The analysis was performed using a field emission SEM (JSM-6460lv, JEOL, Japan) equipped with energy dispersive spectroscopy (EDS) (TEAM Apex XL, EDAX, America). The SEM was operated at a beam current of 10 nA and an accelerating voltage of 25 kV to observe the textural characteristics of the samples and to perform EDS analysis.

4.3 LA-ICP-MS analysis

The in-situ trace element composition of individual sphalerite grains was analyzed using LA-ICP-MS at the SKLODG, IGCAS. Laser sampling was performed using an ASI RESOLution-LR-S155 laser microprobe with a Coherent Compex-Pro 193 nm ArF excimer laser. Ion-signal intensities were measured using an Agilent 7700x ICP-MS instrument. Helium (350 ml/min) was used as a carrier gas, and the ablated aerosol was mixed with argon (900 ml/min) as a transport gas before exiting the cell. Each analysis included approximately 20 seconds of background acquisition (gas blank) followed by 45 seconds of data acquisition from the sample. The analysis was conducted with a pit size of 26 μm, a pulse frequency of 5 Hz, and a fluence of 3 J/cm². The elements measured included ^{55}Mn, ^{57}Fe, ^{59}Co, ^{60}Ni, ^{65}Cu, ^{66}Zn, ^{71}Ga, ^{72}Ge, ^{75}As, ^{77}Se, ^{107}Ag, ^{111}Cd, ^{115}In, ^{118}Sn, ^{121}Sb, ^{125}Te, ^{205}Tl, ^{208}Pb, and ^{209}Bi. The standard STDGL3 was used to determine concentrations of chalcophile and siderophile elements (Danyushevsky et al., 2011), while in-house standard pure pyrite was used for the calibration of S and Fe concentrations. The integrated count data were calibrated and converted using GSD-1G for lithophile elements, and the sulfide reference material GSE-1G was analyzed as an unknown sample to check the analytical accuracy.

5. Results

5.1 Petrography

The ores from the Shanshulin deposit exhibit massive (Fig. 3C, D), veined (Fig. 3F, G), and disseminated (Fig. 3E) structures, along with anhedral to euhedral granular (Fig.4 A, B), zonal (Fig. 4D), filling (Fig. 4J, L), megacryst (Fig. 3H), and fine-grained textures (Fig. 4B). Pyrite is euhedral to subhedral with a particle size of 0.01-0.1 mm and is disseminated in dolomite (Fig. 4B), whereas pyrite associated with sphalerite generally has a particle size ranging from 0.05 mm to 0.8 mm and often forms aggregates (Fig. 4C, H). Cracks in pyrite are frequently filled with galena and calcite (Fig. 4J, K, L). The surface of sphalerite (Sp1) contains mineral inclusions such as pyrite and quartz (Fig. 4H, I). Based on the color of Sp1, it is further classified into two generations: Sp1a and Sp1b. Sp1a is dark brown, with many cracks and fine-grained pyrite and quartz inclusions (Fig. 4A, D). Sp1b is light brown, with fewer inclusions and typically grows around Sp1a, forming a zoning texture with different colors (Fig. 4B, D). The surface of sphalerite (Sp2) lacks mineral inclusions and is light yellow to white (Fig. 4C, F). Calcite often crosscuts the surrounding rock and ore bodies (Fig. 4B, E). The boundaries between calcite and Sp2 are typically smooth (Fig. 4C). Based on the mineral assemblages and crosscutting relationship between minerals, the paregenesis of the Shanshulin deposit was established (Fig. 5).

5.2 Trace elements in sphalerite

Table 1 summarizes the trace elements concentrations in Sp1a (n = 88), Sp1b (n = 73), and Sp2 (n = 38), including average, median, maximum and minimum values, with detailed information provided in Supplementary Table 1. The changes in element concentrations across different stages are shown in Fig. 6.

A significant decline in the concentrations of Mn and Fe is observed from the early to late stage. The average concentration of Mn decreases from 113 ppm in Sp1a to 70 ppm in Sp1b and then to 21 ppm in Sp2. The average concentration of Fe exhibits a similar pattern, dropping from 23461 ppm in Sp1a to 12048 ppm in Sp1b and then to 5310 ppm in Sp2.

The average concentration of Cd increases from 765 ppm in Sp1a to 878 ppm in Sp1b and then to 1087 ppm in Sp2. Cu, Ge, and Ag follow a pattern of initially increasing and then decreasing. The concentration of Cu is generally low. Its average concentration is 57 ppm in Sp1a, 96 ppm in Sp1b, and 81 ppm in Sp2. Ge is enriched in the sphalerite of the Shanshulin deposit. Its average concentration is 169 ppm in Sp1a, 210 ppm in Sp1b, and 126 ppm in Sp2. From the early stage to the late stage, the average concentration of Ag is 100 ppm in Sp1a, 146 ppm in Sp1b, and 77 ppm in Sp2.

The average Ga concentration is 9.3 ppm in Sp1a, 20.8 ppm in Sp1b, and 13.1 ppm in Sp2. Only a few analyses of Ga exceed 100 ppm. The average concentration of Pb is 46 ppm in Sp1a, 66.5 ppm in Sp1b, and 7.9 ppm in Sp2. For Sn, the average concentration is 2.0 ppm in Sp1a, 10.6 ppm in Sp1b, and 14.6 ppm in Sp2. The average concentration of Sb is 6.5 ppm in Sp1a, 4.9 ppm in Sp1b, and 2.9 ppm in Sp2. As, Co, Ni, and In have low concentrations, mostly below 1 ppm, and many analyses are below the detection limit.

In summary, Sp1a is relatively rich in Fe and Mn but low in Cu and Cd. Germanium and Ag are relatively enriched in both Sp1a and Sp1b. Sp2 is relatively rich in Cd but has lower contents of Fe, Mn, Ge, and Ag. The comprehensive recovery and utilization grade of Ge in zinc concentrate is 0.01%, so the Ge content in the three types of sphalerites in Shanshulin meets the requirements for industrial utilization. Gallium is not significantly enriched in the Shanshulin deposit.

6. Discussion

6.1 The occurrence and substitution mechanism of Ge

The LA-ICP-MS time-resolved depth-intensity profile can provide information about the occurrence of specific elements within sphalerite, with a smooth profile indicating lattice substitution and a peak profile indicating the presence of mineral inclusions (Cook et al., 2009). Elements such as Fe, Ge, Ag and Pb exhibit smooth profiles (Fig. 7A), suggesting that they are present in the lattice of sphalerite as isomorphism. In Fig.7B, Pb and Sb show spike profiles, which are generally indicative of sulfosalt or galena inclusions within sphalerite (Cook et al., 2009). In Fig. 7C, the synchronous spike profiles of Ge and Ag suggest the possible presence of Ge-and-Ag-containing mineral inclusions. Research has shown that a high Ag environment is conducive to Ge sulfides formation (Bernstein, 1985). The average concentration of Ag in the Shanshulin deposit is 108 ppm, which indicates that there is indeed potential for the formation of Ge sulfides or nanoparticles in the Shanshulin deposit. However, the variation in the concentration of Ge near the crystal-fluid interface may also lead to spike or not smooth profiles in the signal feedback of LA-ICP-MS time-resolved profile (Wu et al., 2019; Sun et al., 2024). Therefore, more detailed methods are needed to confirm the presence of Ge-rich nanoparticles in the Shanshulin deposit (Wu et al., 2019).

The substitution mechanism of Ge into sphalerite is affected by factors such as the nature of ore-forming fluids (e.g., Ag or Cu concentration) and the crystal field stabilization energy of elements (Belissont et al., 2014). Therefore, the substitution mechanism of Ge into sphalerite varies across different deposits, and even among different generations or types of sphalerites within the same deposit ((Yuan et al., 2018; Wu et al., 2019; Bai et al., 2024). Studies have shown that Ge enters sphalerite mainly through directly replacing zinc or coupling with elements such as Cu^{+}, Ag^{+}, Fe^{2+}, and Pb^{2+} to replace zinc (Cook et al., 2015; Belissont, 2016; Cugerone et al., 2021; Bai et al., 2024; Cheng et al., 2024). The (Ag +Cu) / Ge molar ratio of sphalerite from the Shanshulin deposit is mostly less than 2 (Fig. 8B), indicating that the $Ge^{4+}+2(Ag^{+}, Cu^{+}) \leftrightarrow 3Zn^{2+}$ substitution cannot account for all the forms of Ge in sphalerite. Therefore, there may be other substitution mechanisms. However, apart from the correlation between Ge and (Ag +Cu), there is no obvious correlation between Ge and elements such as Fe and Pb (Fig. 8A, C), which suggests that Ge in sphalerite with $(Ag/Ge)_{mol} < 2$ may be incorporated through the direct substitution of $Ge^{2+} \leftrightarrow Zn^{2+}$ or $Ge^{4+} + \square$ ($\square$

represents vacancy) ↔ $2Zn^{2+}$. A strong negative correlation exists between Zn and Fe, Mn (Fig. 8D), and the ionic radii of Fe^{2+} (0.63 Å) and Mn^{2+} (0.66 Å) are similar to that of Zn^{2+} (0.60 Å), allowing Fe^{2+} and Mn^{2+} to be incorporated into sphalerite through the (Fe^{2+}, Mn^{2+}) ↔ Zn^{2+} substitution. Additionally, a weakly negative correlation between Fe and Cd (Fig. 8E), and a weakly positive correlation between Cd and Zn (Fig. 8F) implies that Fe and Cd will compete each other during partition into sphalerite. In the fluids with a high Fe concentration, the incorporation of Cd into sphalerite will be hindered.

6.2 The influence of temperature and fS_2 on enrichment of trace elements

Trace element composition of sphalerite can be used to explore the physical and chemical changes of hydrothermal fluids such as temperature and sulfur fugacity during sphalerite precipitation (Frenzel et al., 2016; Fontboté et al., 2017; Keith et al., 2018; Meng et al., 2019; Salvioli-Mariani et al., 2024). According to the GGIMFis geothermometer proposed by Frenzel et al. (2016), which is expressed as $T(℃) = -(54.4 \pm 7.3) \cdot PC1^* + (208 \pm 10)$, with $PC1^* = ln\left(\frac{c_{Ga}^{0.22} \cdot c_{Ge}^{0.22}}{c_{Fe}^{0.37} \cdot c_{Mn}^{0.20} \cdot c_{In}^{0.10}}\right)$, the mineralization temperature of the Shanshulin deposit was calculated. In combination with the *FeS*-activities, the fS_2 value was also calculated (Frenzel et al., 2022). This method is not only applicable to large-scale but also to small-scale mineralization events (Frenzel et al., 2022). The premise for estimating fS_2 is that the Fe concentration in sphalerite comes from pyrite buffering. At different stages of Pb-Zn mineralization in the Shanshulin deposit, pyrite is commonly present, which is replaced, intercalated, or encapsulated by sphalerite. Therefore, this assumption is reasonable for the Shanshulin deposit. The specific calculation process of fS_2 can be found in Frenzel et al. (2022).

The GGIMFis temperature mainly ranges between 125-232℃ (Fig. 9), consistent with the previously reported sphalerite fluid inclusions temperature of 115-207℃ (Yang et al., 2024). Different generations of sphalerite falls within the intermediate-sulfidation fields (Fig. 10). Because a low-sulfur environment promotes Ge to more readily enter the sphalerite lattice (Bernstein, 1985), the intermediate fS_2 may hinder Ge entering the sphalerite lattice and in contrast, Ge has the potential to form Ge sulfides (Fig. 7C). Furthermore, the evolutionary trend of the sphalerite generations follows the "S-gas buffer", which indicates that the chemical reactions and mass exchanges between gaseous sulfur species such as H_2S or SO_2 and minerals or fluids may be the main driver for the changes in temperature and fS_2 (Einaudi et al., 2003).

Decline in temperature hinders the enrichment of Mn and Fe, promotes the enrichment of Cu and Ga, and has little effect on Ge and Ag (Fig. 11). However, the research results of Frenzel et al. (2016) indicate that Ge is closely related to temperature. This discrepancy might be attributed to the wider temperature range of the deposits they studied, making the controlling effect of temperature on Ge more prominent. Iron has a correlation with dfS_2, while other trace elements lack correlation with dfS_2 (Fig. 12). Since the calculation of fS_2 depends on the Fe content in sphalerite, fS_2 and Fe content are inherently connected. Consequently, we conclude that the controlling effect of fS_2 on the enrichment of elements in sphalerite from the Shanshulin deposit is not significant. Based on the above analysis, we speculate that Ge is jointly regulated by the temperature and composition of the ore-forming fluids. We propose a model to depict the differential enrichment of trace elements in sphalerite from the Shanshulin deposit (Fig. 13). In the stage for Sp1a formation, the relatively high ore-forming temperature not only promotes the preferential partition of Fe and Mn into sphalerite but also inhibits the partition of Cu and Ga. Consequently, in the ore-forming fluid, the concentrations of Fe and Mn decrease, and those of Cu and Ga increase. Since Fe and Mn are the "colored elements", their enrichment makes Sp1a appear dark brown. On the other hand, the relatively high temperature impedes the partition of Cu into sphalerite, thus hindering the incorporation of Ge via $Ge^{4+}+2Cu^{+} \leftrightarrow 3Zn^{2+}$. Therefore, Ge enters Sp1a mainly through $Ge^{4+}+2Ag^{+} \leftrightarrow 3Zn^{2+}$ and $Ge^{2+} \leftrightarrow Zn^{2+}$ or $Ge^{4+} + \square \leftrightarrow 2Zn^{2+}$. In the stage for Sp1b formation, both the ore-forming temperature and the concentrations of Fe and Mn in the ore-forming fluids decrease. This leads to a reduction in the amount of Fe and Mn entering Sp1b, thus making Sp1b appear light brown. Moreover, the decreased temperature and increased Cu concentration in the ore-forming fluid facilitate Ge to enter sphalerite via $Ge^{4+}+2Cu^{+} \leftrightarrow 3Zn^{2+}$, $Ge^{4+}+2Ag^{+} \leftrightarrow 3Zn^{2+}$, $Ge^{2+} \leftrightarrow Zn^{2+}$ or $Ge^{4+} + \square \leftrightarrow 2Zn^{2+}$. Therefore, multiple substitution mechanisms improve the enrichment efficiency of Ge, enabling the average Ge content in Sp1b to reach the highest level (Fig. 6). In the stage for Sp2 formation, the decrease in the temperature and the concentrations of elements such as Fe, Mn, Cu, Ge, and Ga leads to fewer Fe and Mn entering Sp2, resulting in Sp2 appearing white. Meanwhile, due to the low concentrations of Cu, Ag, and Ge in the ore-forming fluid, only a small amount of Ge can be enriched in Sp2.

6.3 Controlling factors of Ge in different Pb-Zn deposits

We collected LA-ICP-MS data from the Tianqiao, Yadu, and Shaojiwan Pb-Zn deposits in the Northwest Guizhou, which are controlled by secondary faults related to the NW-trending Ziyun-Yadu major fault. The average concentrations of Ge in sphalerite from the Shanshulin, Shaojiwan, Yadu, and Tianqiao deposits are 168 ppm, 111 ppm, 83 ppm, and 80 ppm, respectively (Appendix Table 3). The average temperatures of sphalerite in these deposits are 153℃, 152℃, 176℃, and 193℃ respectively. There is a correlation between the concentration of Ge and temperature in these deposits. In a specific deposit, temperature controls the enrichment of Ge by influencing the substitution mechanism of $Ge^{4+} + 2Cu^{+} \leftrightarrow 3Zn^{2+}$. For example, there is a strong correlation between Cu and Ge in the Tianqiao and Yadu deposits (Fig. 15C, F), indicating that Ge mainly enters sphalerite through the substitution mechanism of $Ge^{4+}+2Cu^{+} \leftrightarrow 3Zn^{2+}$ in these two deposits. Since Cu is controlled by temperature (Fig.

11C), there is an obvious correlation between Ge and temperature (Fig. 15A, D). Most analyses of the Shaojiwan and Shanshulin deposits have $(Cu/Ge)_{mol}$ ratios less than 2 (Fig. 15I, L), suggesting the existence of other substitution mechanisms besides the mechanism $Ge^{4+} + 2Cu^{+} \leftrightarrow 3Zn^{2+}$. Temperature can only affect part of Ge entering sphalerite through $Ge^{4+} + 2Cu^{+} \leftrightarrow 3Zn^{2+}$. Therefore, there is no obvious correlation between Ge and temperature (Fig. 15G, J). In addition, a correlation between Ge and dfS_2 in Yadu and Shaojiwan deposits (Fig. 15E, H) indicates that they may be controlled by fS_2, which needs further research.

7. Conclusion

We investigated the occurrence and substitution mechanism of Ge in the Shanshulin deposit, and identified the key factors influencing Ge enrichment. Germanium may exist in sphalerite in the form of isomorphism and nanoparticles. The substitution mechanisms of Ge likely include $Ge^{4+}+2(Ag^{+}, Cu^{+}) \leftrightarrow 3Zn^{2+}$ and $Ge^{2+} \leftrightarrow Zn^{2+}$ or $Ge^{4+} + \square$ ($\square$ represents vacancy) $\leftrightarrow 2Zn^{2+}$. The S-gas buffer may induce the changes of the temperature and fS_2 of ore-forming fluids. The concentration of Fe, Mn, Cu, and Ga in sphalerite is influenced by temperature, whereas the concentration of Ge is jointly controlled by the temperature and composition of the ore-forming fluids. The difference in Ge enrichment among different deposits in northwestern Guizhou is controlled by temperature which influences the substitution mechanism of Ge.

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Figure captions

Fig. 1. (A) Regional tectonic geological setting of Sichuan-Yunnan-Guizhou (SYG) metallogenic province; (B) Geologic map of the SYG metallogenic province showing the distribution of Pb-Zn deposits, Emeishan basalts and faults (modified from Zhou et al. 2014).

Fig. 2. (A) Geological map of the Shanshulin Pb–Zn deposit showing the faults, folds and projection of the ore body (modified from Yang et al. 2018). (B) A–B cross-section of the Shanshulin Pb–Zn deposit showing the ore bodies and ore-controlling faults (modified from Wu et al. 2023).

Fig. 3. Sulfide mineralization section and photos of ore samples in the Shanshulin deposit. (A) Layered ore body, with a small amount of galena and calcite on its edge. (B) Layered ore bodies are interspersed by veined ore bodies. (C) Massive sphalerite ore containing a small amount of galena. (D) Massive sphalerite ore, with calcite on the edge of the ore. (E) Disseminated ore. (F) Vein sphalerite ore, light-colored sphalerite and calcite form veinlets. (G) Network vein ore, a small amount of light-colored sphalerite and calcite are symbiotic. (H) Megacryst sphalerite particles are enclosed by calcite. Mineral abbreviations: Py = pyrite, Sp = sphalerite, Gn = galena, Dol = dolomite, Cal = calcite.

Fig. 4. Representative photographs of mineralization characteristics of the Shanshulin deposit. (A) Sphalerite is dissolved and replaced by calcite into anhedral granular texture (reflected light). (B) Pyrite is distributed in dolomite, while calcite veins cut through the surrounding rock and sphalerite (reflected light). (C) The boundary between calcite and sphalerite is very flat, and pyrite is enclosed by sphalerite (reflected light). (D) Sphalerite has a zonal texture, usually the core is dark brown and the edge is light brown (transmitted light). (E) Sphalerite is light brown, and calcite veins penetrate the surrounding rock and sphalerite (transmitted light). (F) The boundary between calcite and sphalerite is very flat, and the sphalerite is white (transmitted light). (G) Sphalerite is closely symbiotic with galena. (reflected light). (H) Cubic pyrite and some quartz and calcite particles are distributed in the gap of sphalerite particles. (I) Fine-grained pyrite particles are wrapped by sphalerite, and sphalerite is partially interspersed with fine-vein calcite (BSE). (J) Calcite and galena are filled in the gap between cubic pyrite and sphalerite (BSE). (K) The euhedral granular Pyrite and quartz grains are distributed in aggregates, and the gaps between the grains are filled with calcite (BSE). (L) Cracks in pyrite are filled with calcite and galena (BSE). Mineral abbreviations: Py = pyrite, Sp = sphalerite, Gn = galena, Dol = dolomite, Cal = calcite, Qtz = quartz.

Fig. 5. Metallogenic stage of Shanshulin Pb-Zn deposit.

Fig. 6. Box and whisker plots of the trace element results of Sp1a, Sp1b, and Sp2 from the Shanshulin deposit. Data below the limit of detection were not considered in this plot.

Fig. 7. Typical LA-ICP-MS time-resolved depth-intensity profiles of sphalerite from the Shanshulin Pb-Zn deposit.

Fig. 8. Correlations between elemental concentrations in different types of sphalerite.

Fig. 9. Box and whisker plots show the variation of temperature in sphalerite at different stages. Note: When calculating the GGIMFis temperature, the detection limits were used to replace the missing values of elements such as Ga and In.

Fig. 10. Sulfur fugacity vs. temperature plot, showing the location of sphalerite relative to sulfur gas, rock buffer, and different mineral reaction lines at different stages of the Shanshulin Pb-Zn deposit (modified from Einaudi et al., 2003). Note: When calculating the fS_2, the pressure of the deposit was obtained from Fig. 14.

Fig. 11. Trace element concentrations in sphalerite as a function of GGIMFis temperature.

Fig. 12. Trace element concentrations in sphalerite as a function of dfS_2. Note: During the evolution of ore-forming fluids, temperature has an impact on fS_2. Therefore, before studying the effect of fS_2 on the composition of sphalerite, we used a linear regression method to eliminate the controlling effect of temperature on fS_2.

Fig. 13. A model for the differential enrichment of trace elements in the sphalerite of the Shanshulin deposit. Note: The curves of temperature, Fe, Mn, and Cu in the figure only represent their variation trends in the ore-forming fluids. The larger the diameter of the sphere is, the greater the quantity of the elements incorporated into the sphalerite will be.

Fig. 14. Temperature-density diagram for NaCl-H_2O system (Bischoff, 1991). The average ore-forming pressures of the Tianqiao, Yadu, Shaojiwan and Shanshulin deposits are approximately 7.5×10^5 Pa, 7.5×10^5 Pa, 5×10^5 Pa and 5×10^5 Pa respectively, which are used to calculate fS_2 of each deposit (data from Yang et al. 2024).

Fig. 15. The relationships between the concentration of Ge in different deposits and GGIMFis temperature, dfS_2 and Cu. (Data of Tianqiao, Yadu and Shaojiwan deposits are from Yang et al. 2024).

Table 1 Summary of LA-ICP-MS data of sphalerite from the Shanshulin Pb-Zn deposit (data in ppm).

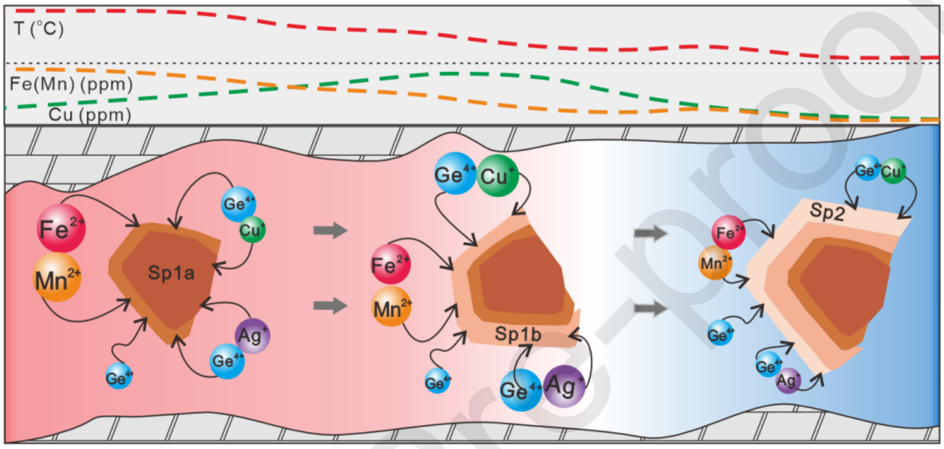
Highlights

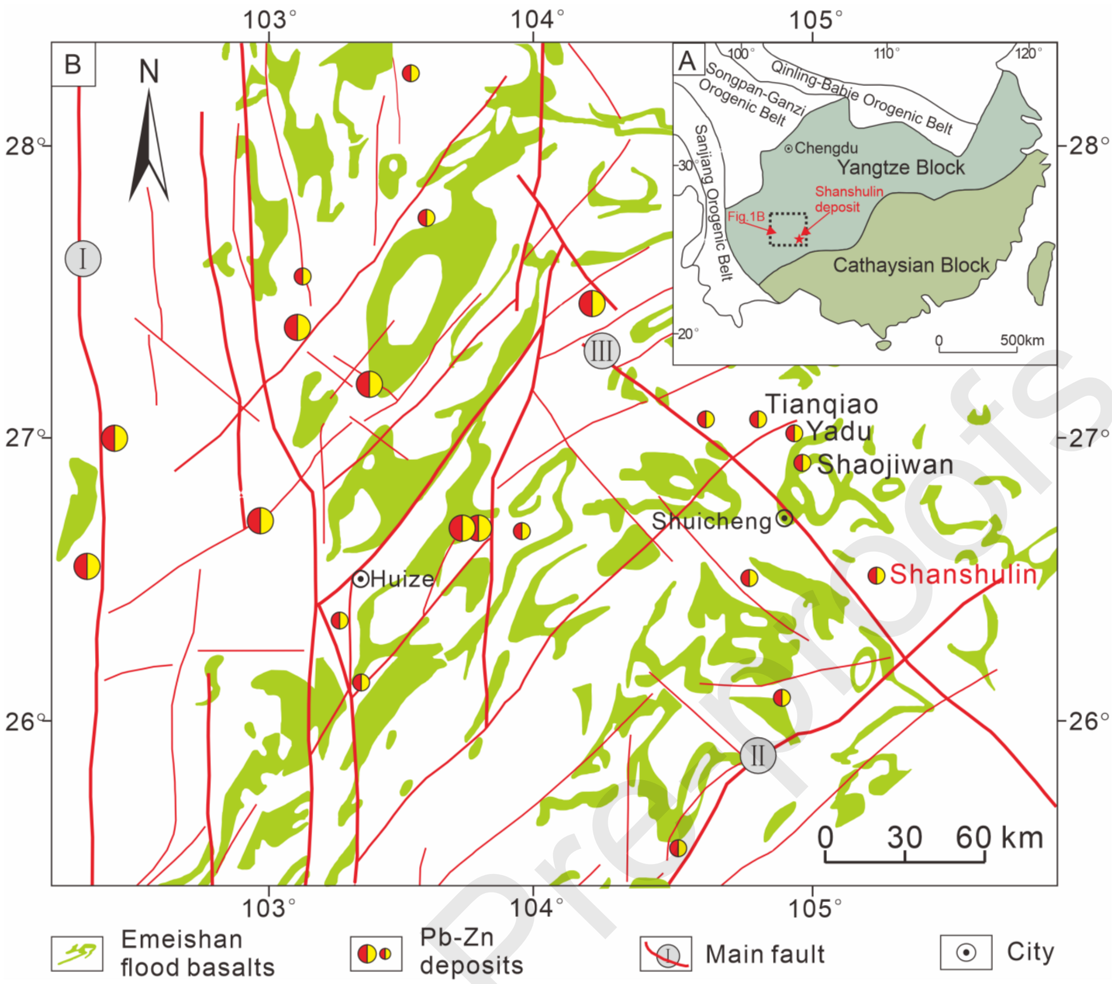
- Ge occurs as isomorphism and nanoparticles.

- The substitution of Ge may be $Ge^{4+}+2(Ag^{+}, Cu^{+}) \leftrightarrow 3Zn^{2+}$ and $Ge^{2+} \leftrightarrow Zn^{2+}$ or $Ge^{4+} + \square \leftrightarrow 2Zn2+$.

- Concentrations of Ge are controlled by the temperature and composition of the ore-forming fluids.

- Temperature influences the substitution mechanisms of Ge.





B
N
103°
104°
105°
28°
27°
26°
A
100°
110°
120°
30°
20°
Songpan-Ganzi Orogenic Belt
Qinling-Babie Orogenic Belt
Sanjiang Orogenic Belt
Chengdu
Yangtze Block
Shanshulin deposit
Fig.1B
Cathaysian Block
0
500km
I
II
III
Tianqiao
Yadu
Shaojiwan
Shuicheng
Shanshulin
Huize
0
30
60 km
Emeishan flood basalts
Pb-Zn deposits
Main fault
City

Figure 1

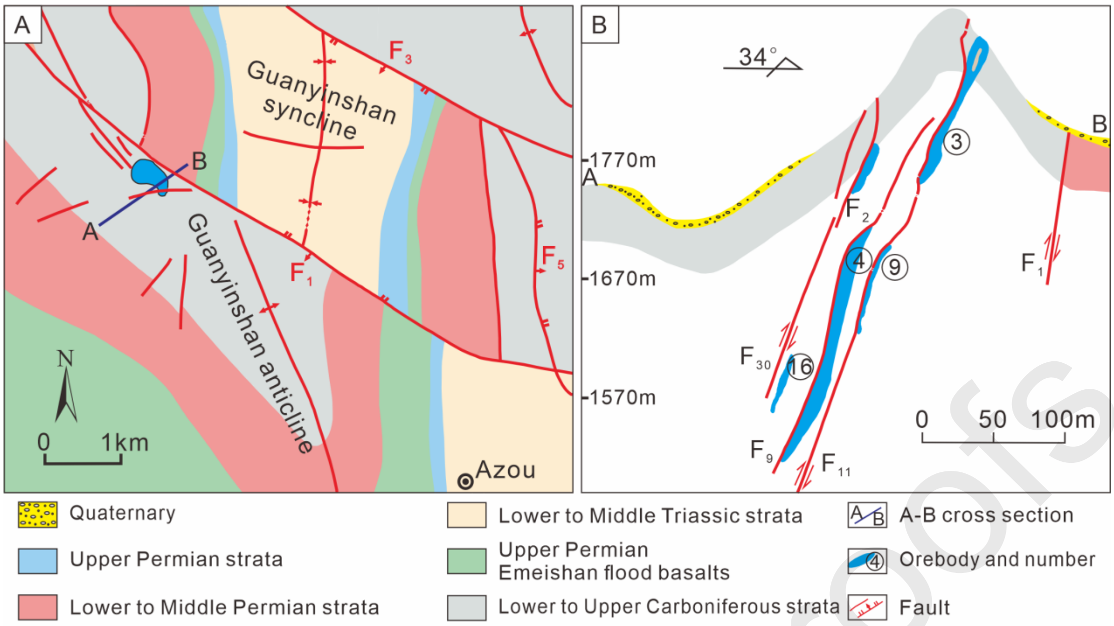

A
Guanyinshan syncline
F3
B
A
Guanyinshan anticline
F1
F5
N
0 1km
Azou
B
34°
1770m
A
1670m
1570m
F2
F1
F30
F9
F11
B
0 50 100m
Quaternary
Upper Permian strata
Lower to Middle Permian strata
Lower to Middle Triassic strata
Upper Permian Emeishan flood basalts
Lower to Upper Carboniferous strata
A-B cross section
Orebody and number
Fault

Figure 2

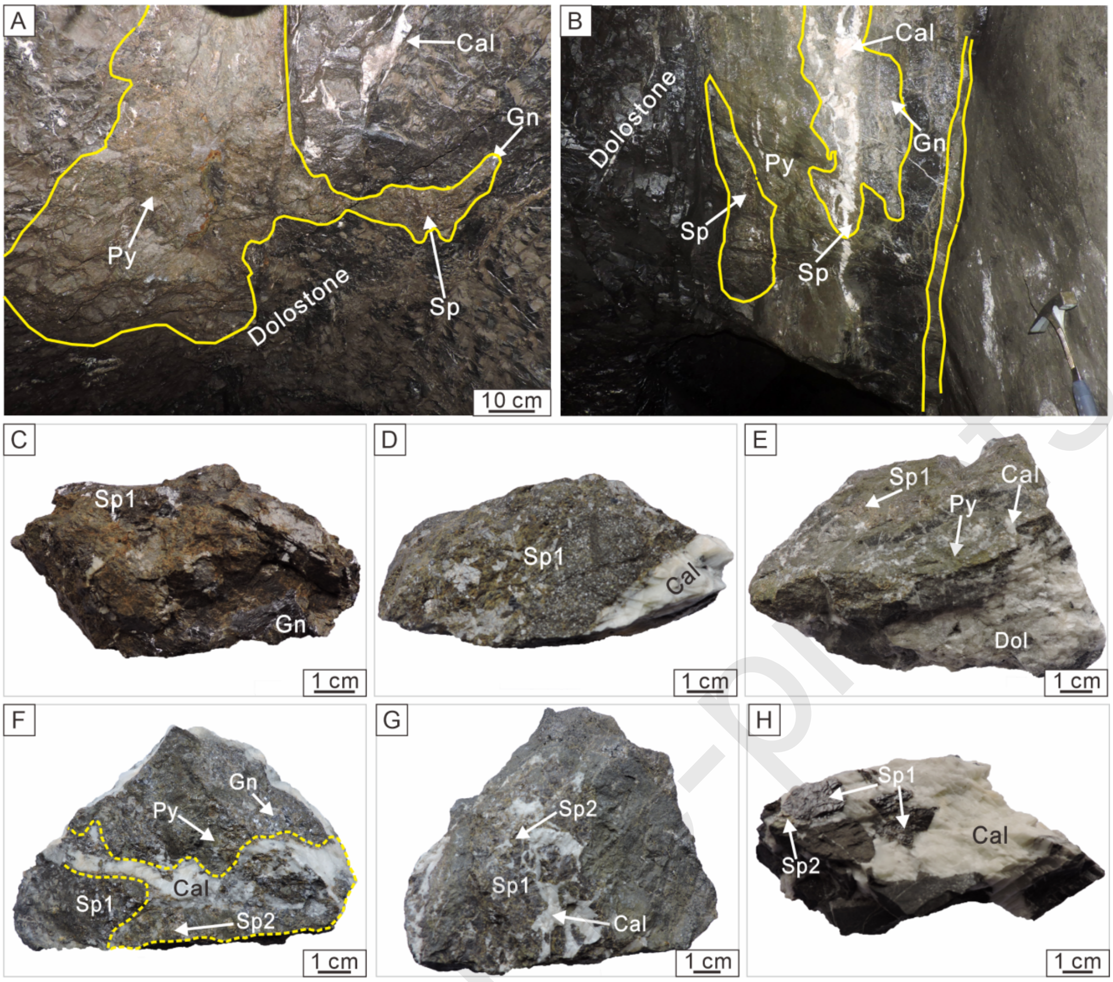

A
Cal
Gn
Py
Sp
Dolostone
10 cm
B
Cal
Dolostone
Py
Gn
Sp
Sp
C
Sp1
Gn
1 cm
D
Sp1
Cal
1 cm
E
Sp1
Cal
Py
Dol
1 cm
F
Gn
Py
Cal
Sp1
Sp2
1 cm
G
Sp2
Sp1
Cal
1 cm
H
Sp1
Sp2
Cal
1 cm

Figure 3

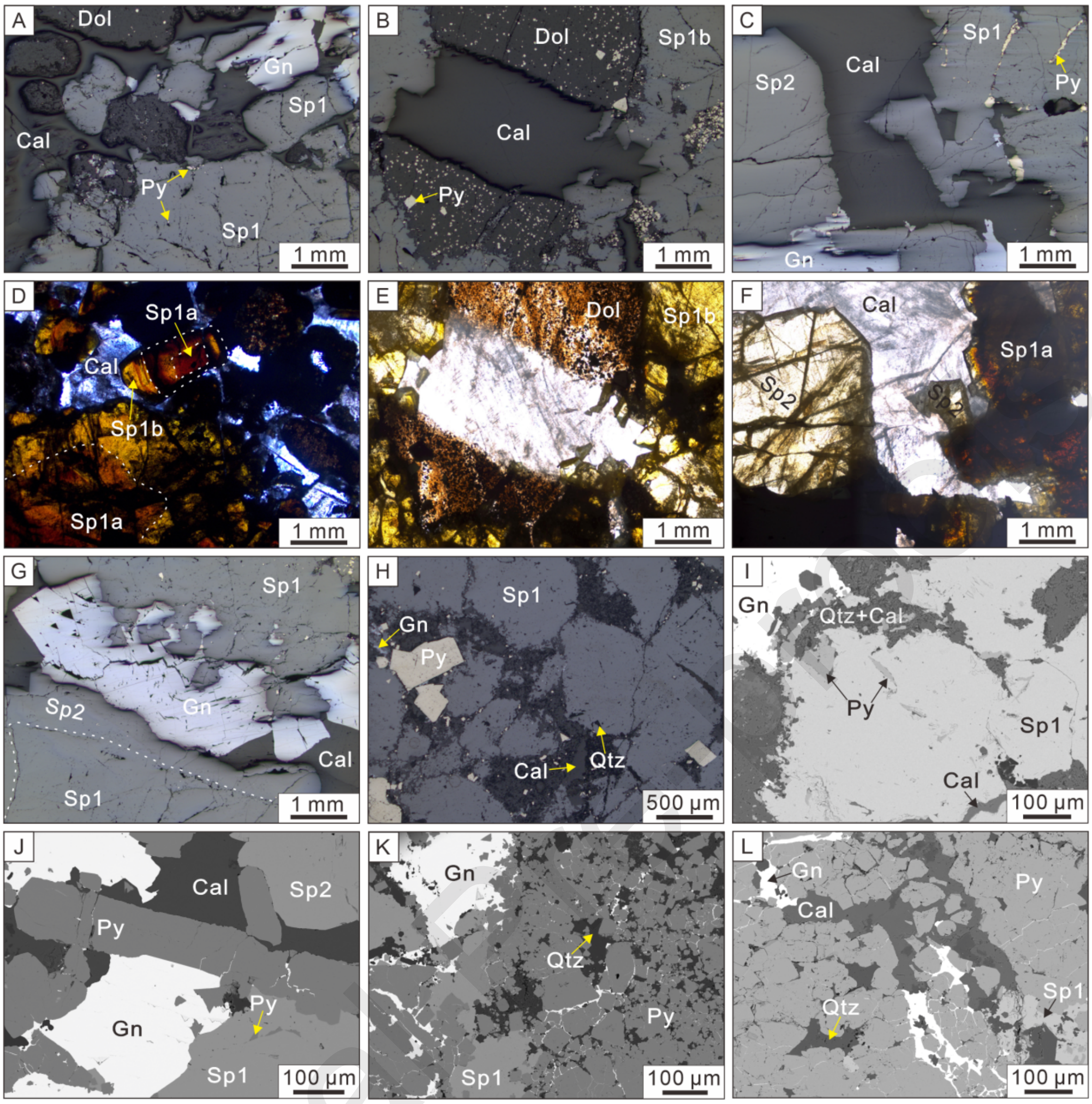

A
Dol
Gn
Sp1
Cal
Py
Sp1
1 mm
B
Dol
Sp1b
Cal
Py
1 mm
C
Sp1
Cal
Sp2
Py
Gn
1 mm
D
Sp1a
Cal
Sp1b
Sp1a
1 mm
E
Dol
Sp1b
1 mm
F
Cal
Sp1a
Sp2
Sp2
1 mm
G
Sp1
Sp2
Gn
Cal
Sp1
1 mm
H
Sp1
Gn
Py
Cal
Qtz
500 μm
I
Gn
Qtz+Cal
Py
Sp1
Cal
100 μm
J
Cal
Sp2
Py
Py
Gn
Sp1
100 μm
K
Gn
Qtz
Py
Sp1
100 μm
L
Gn
Cal
Py
Sp1
Qtz
100 μm

Figure 4

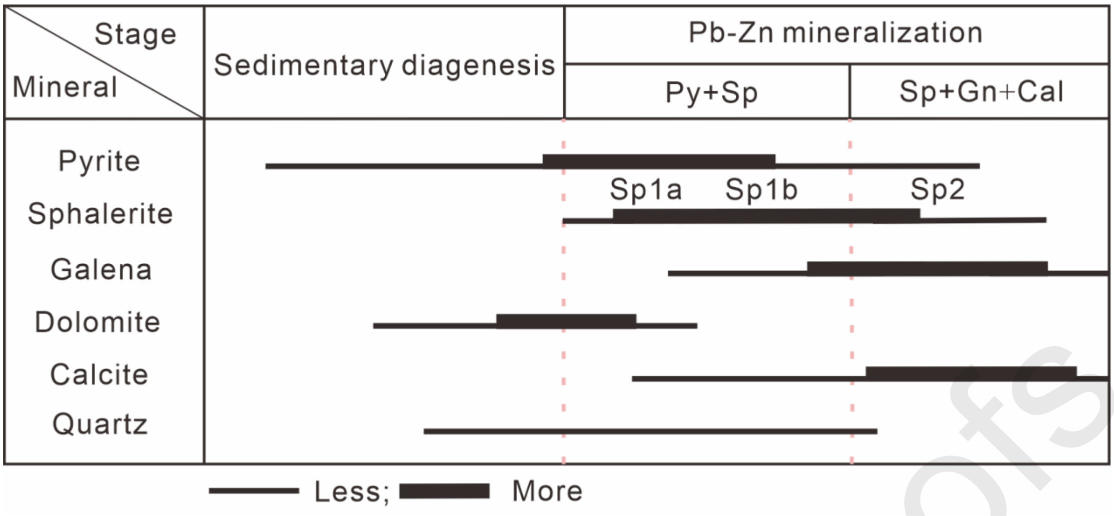
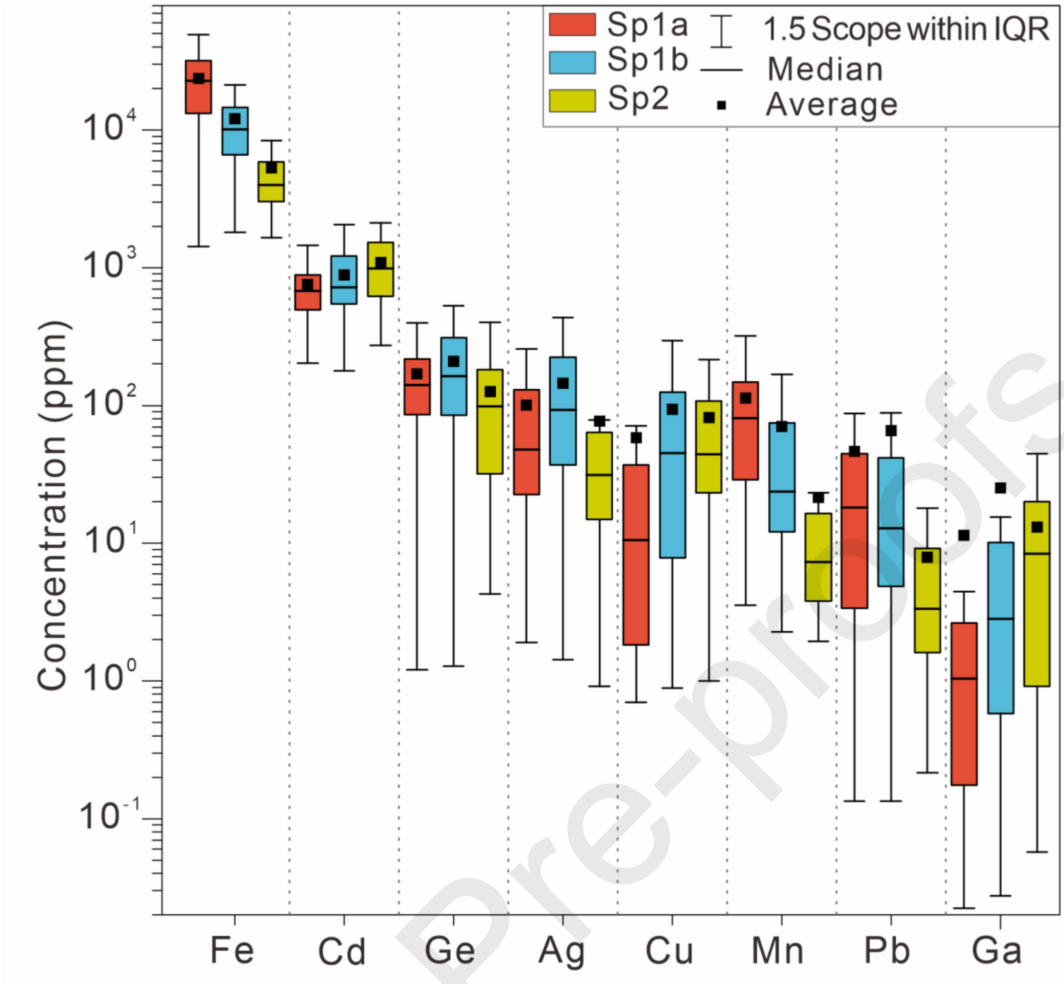

Stage
Mineral
Sedimentary diagenesis
Pb-Zn mineralization
Py+Sp
Sp+Gn+Cal
Pyrite
Sphalerite
Galena
Dolomite
Calcite
Quartz
Sp1a
Sp1b
Sp2
Less;
More

Figure 5



Sp1a
Sp1b
Sp2
1.5 Scope within IQR
Median
Average
Concentration (ppm)
10^4
10^3
10^2
10^1
10^0
10^-1
Fe
Cd
Ge
Ag
Cu
Mn
Pb
Ga

Figure 6

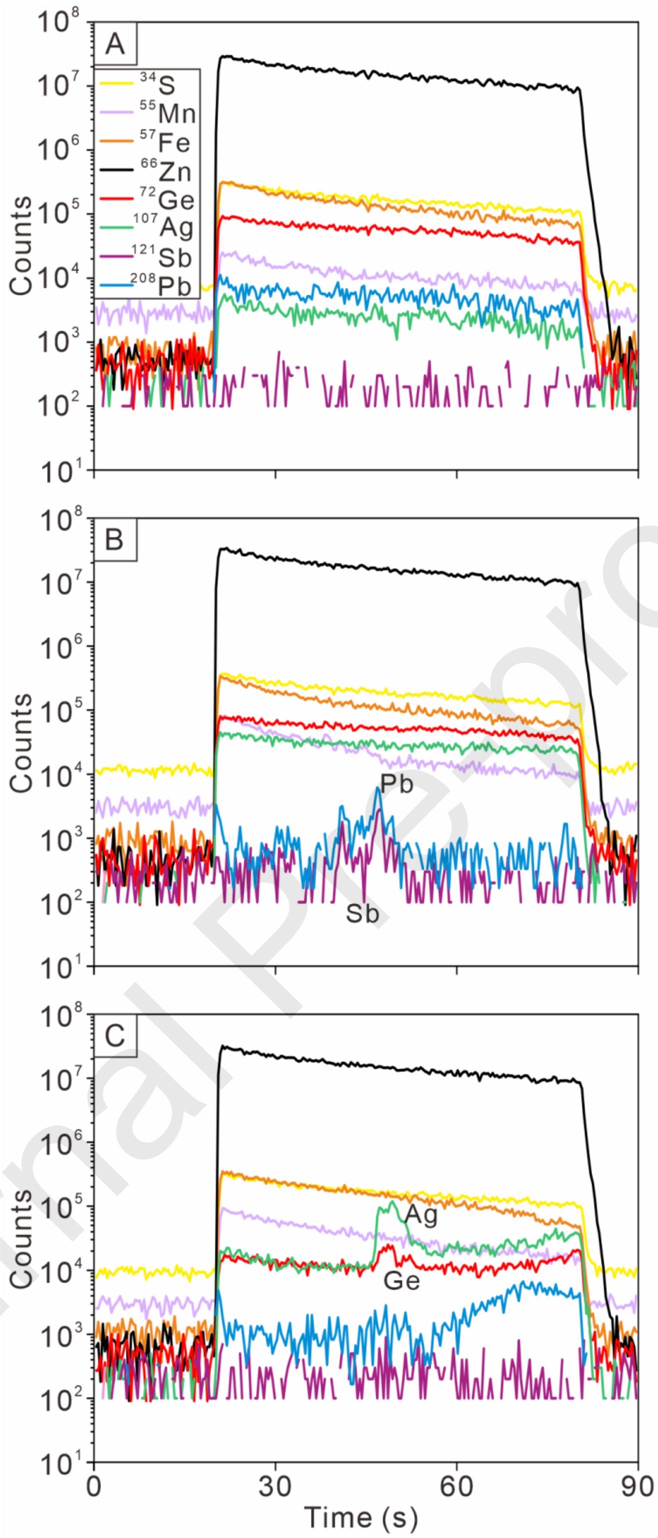

A
B
C
^{34}S
^{55}Mn
^{57}Fe
^{66}Zn
^{72}Ge
^{107}Ag
^{121}Sb
^{208}Pb
Pb
Sb
Ag
Ge
Counts
Time (s)
0
30
60
90
10^1
10^2
10^3
10^4
10^5
10^6
10^7
10^8

Figure 7

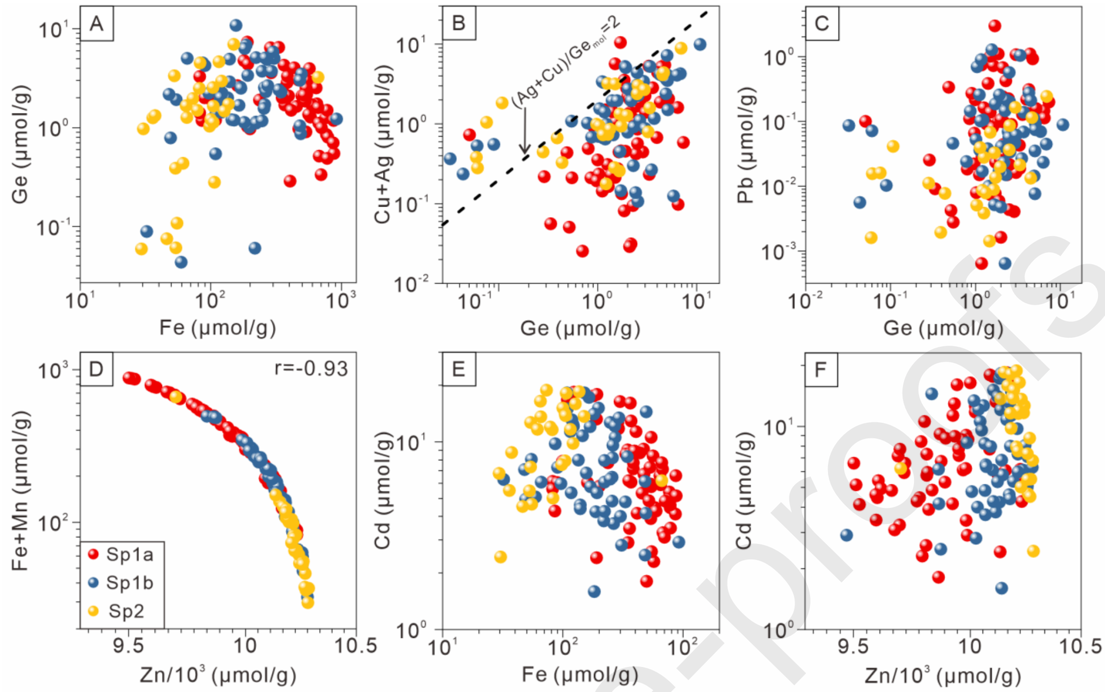
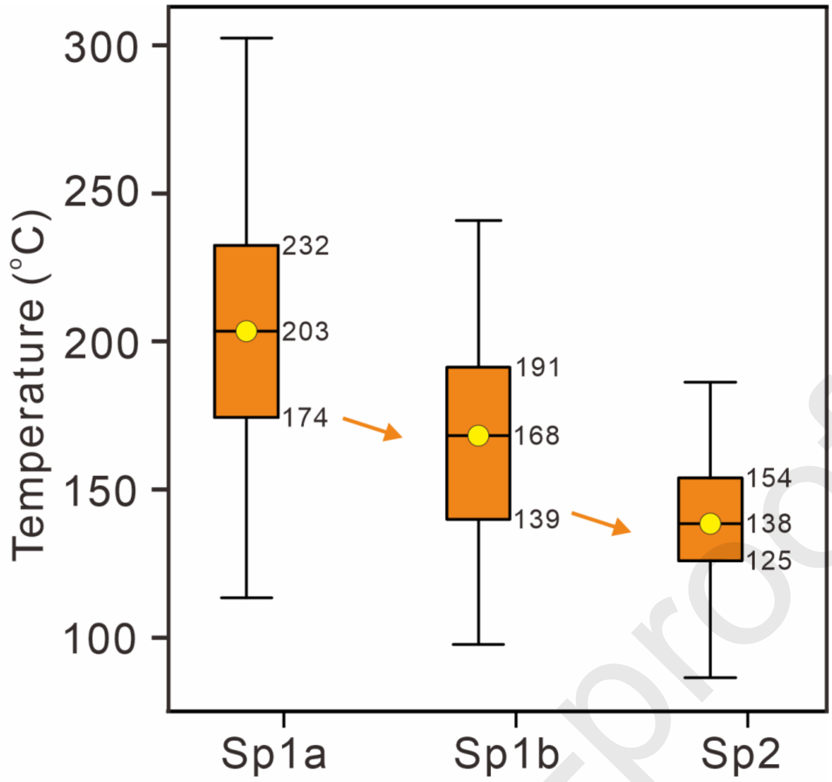

A
Ge (μmol/g)
Fe (μmol/g)
B
(Ag+Cu)/Ge$_{mol}$=2
Cu+Ag (μmol/g)
Ge (μmol/g)
C
Pb (μmol/g)
Ge (μmol/g)
D
r=-0.93
Fe+Mn (μmol/g)
Zn/10^3 (μmol/g)
Sp1a
Sp1b
Sp2
E
Cd (μmol/g)
Fe (μmol/g)
F
Cd (μmol/g)
Zn/10^3 (μmol/g)

Figure 8



Temperature (°C)
300
250
200
150
100
232
203
174
191
168
139
154
138
125
Sp1a
Sp1b
Sp2

Figure 9

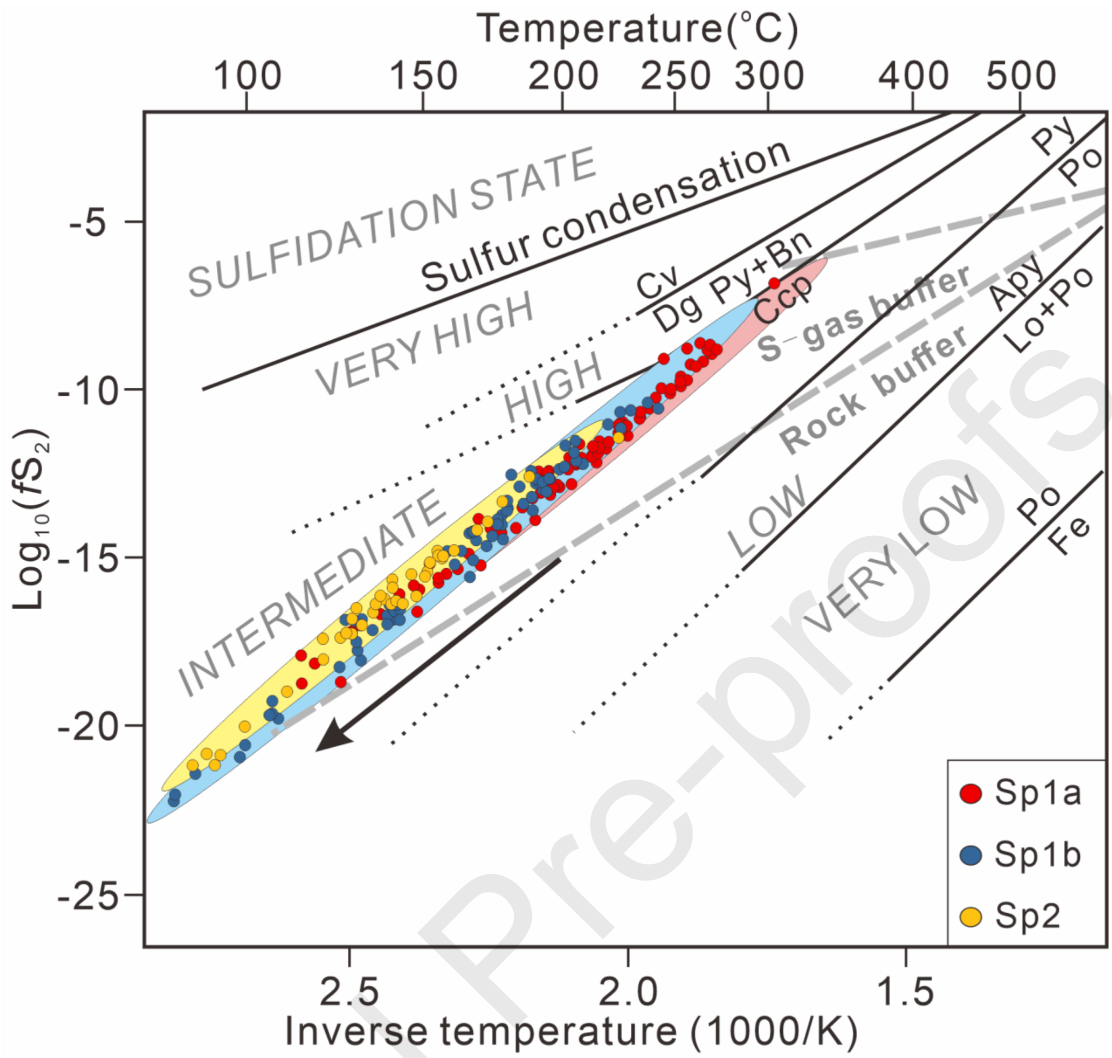

Temperature(°C)
100
150
200
250
300
400
500
SULFIDATION STATE
Sulfur condensation
VERY HIGH
Cv
Dg
Py+Bn
Ccp
HIGH
INTERMEDIATE
S-gas buffer
Rock buffer
Py
Po
Apy
Lo+Po
LOW
VERY LOW
Po
Fe
Log10(fS2)
-5
-10
-15
-20
-25
2.5
2.0
1.5
Inverse temperature (1000/K)
Sp1a
Sp1b
Sp2

Figure 10

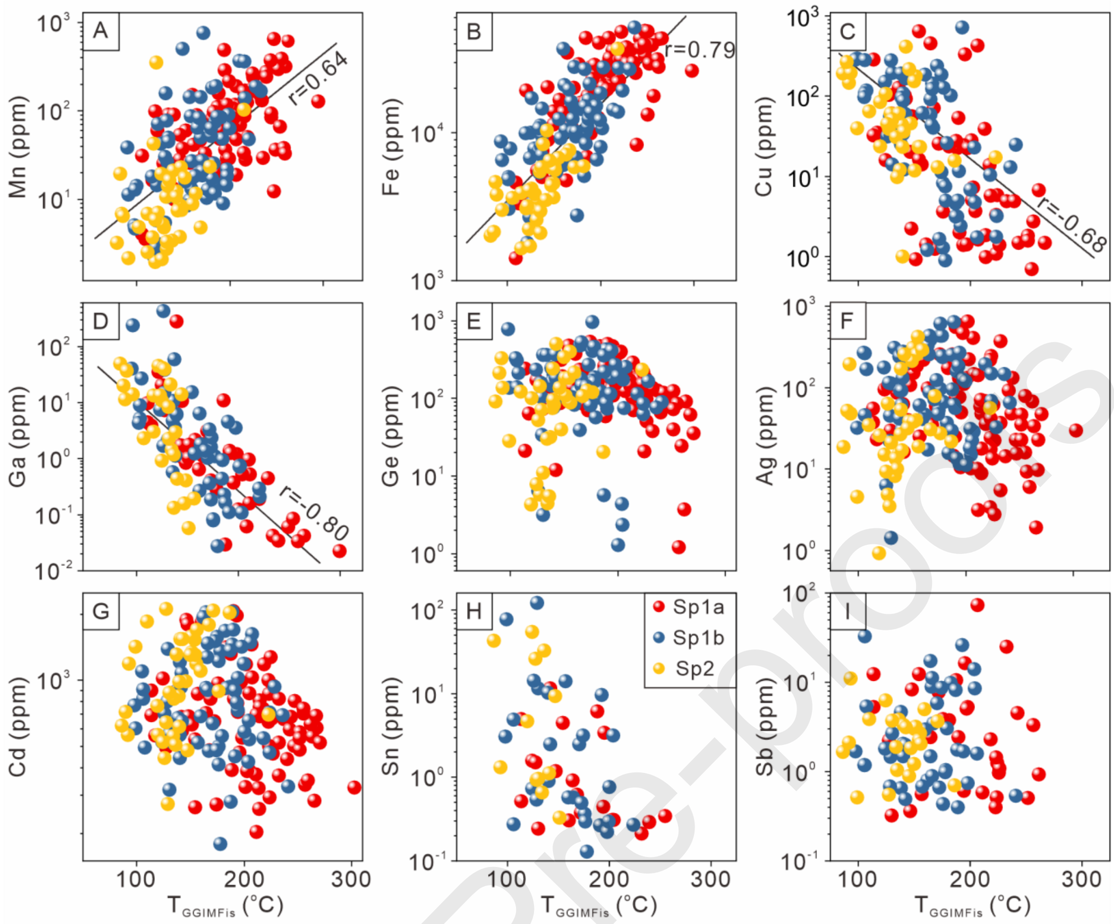

A
B
C
D
E
F
G
H
I
r=0.64
r=0.79
r=-0.68
r=-0.80
Mn (ppm)
Fe (ppm)
Cu (ppm)
Ga (ppm)
Ge (ppm)
Ag (ppm)
Cd (ppm)
Sn (ppm)
Sb (ppm)
Sp1a
Sp1b
Sp2
100
200
300
$T_{GGIMFis}$ (°C)

Figure 11

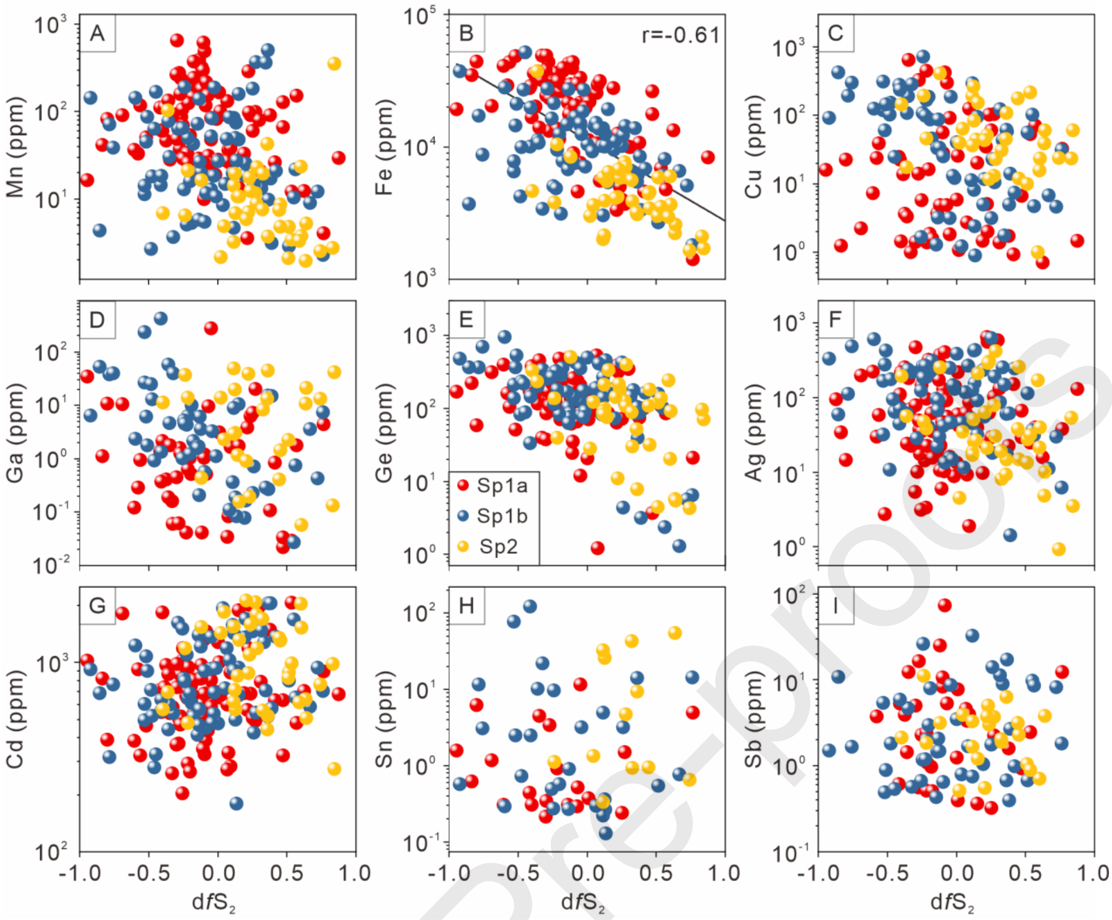

A
B
r=-0.61
C
D
E
F
G
H
I
Mn (ppm)
Fe (ppm)
Cu (ppm)
Ga (ppm)
Ge (ppm)
Ag (ppm)
Cd (ppm)
Sn (ppm)
Sb (ppm)
Sp1a
Sp1b
Sp2
dfS2

Figure 12

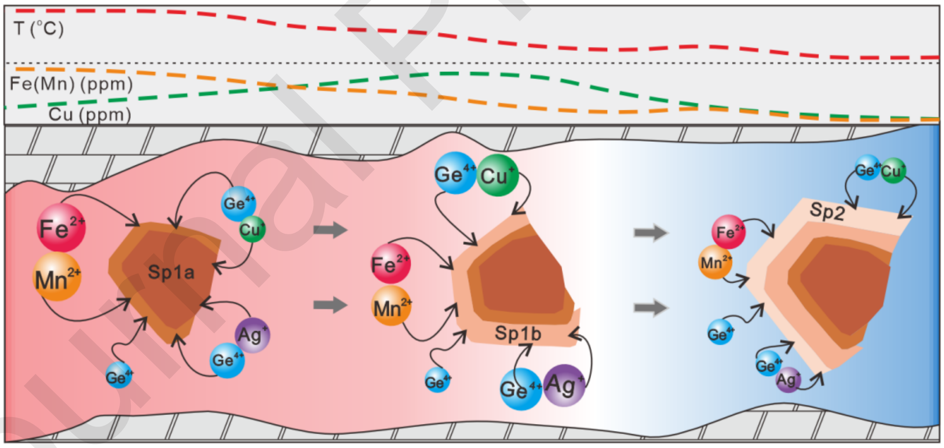


Figure 13

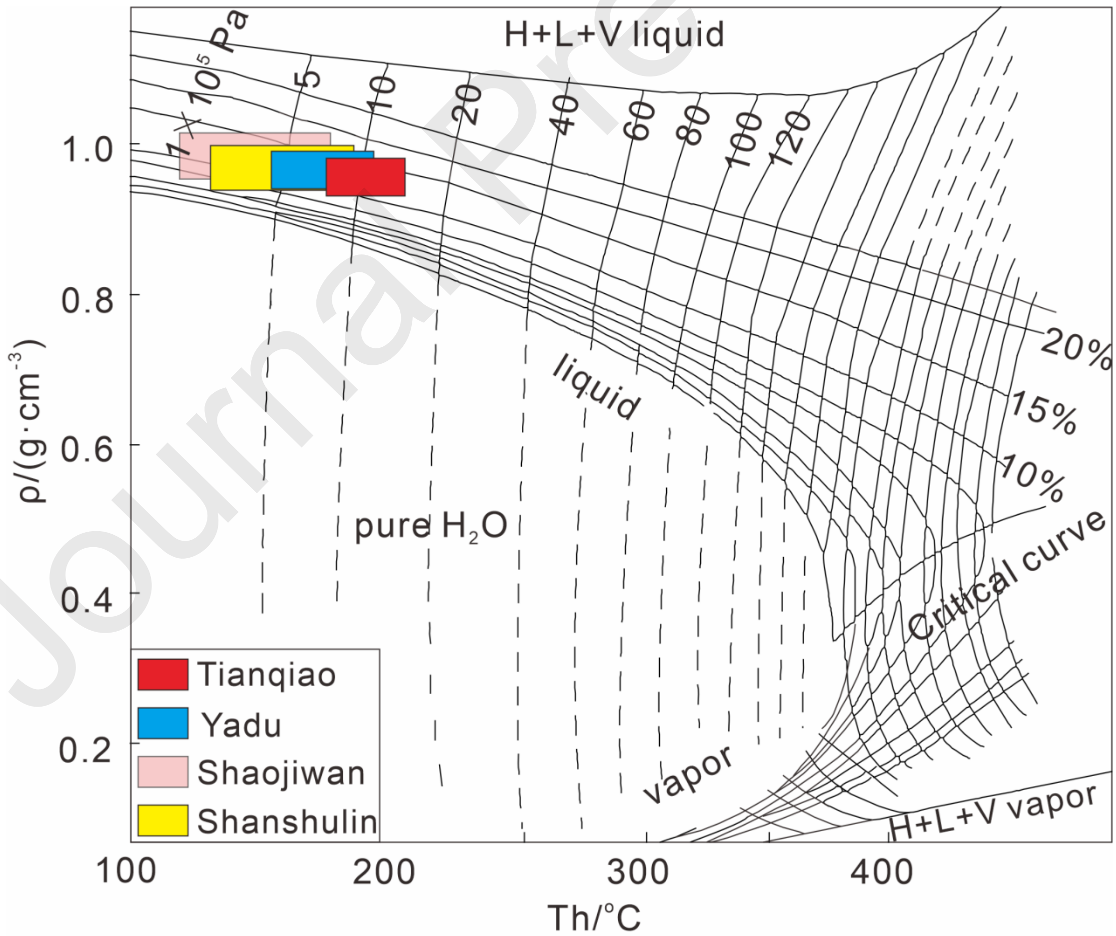
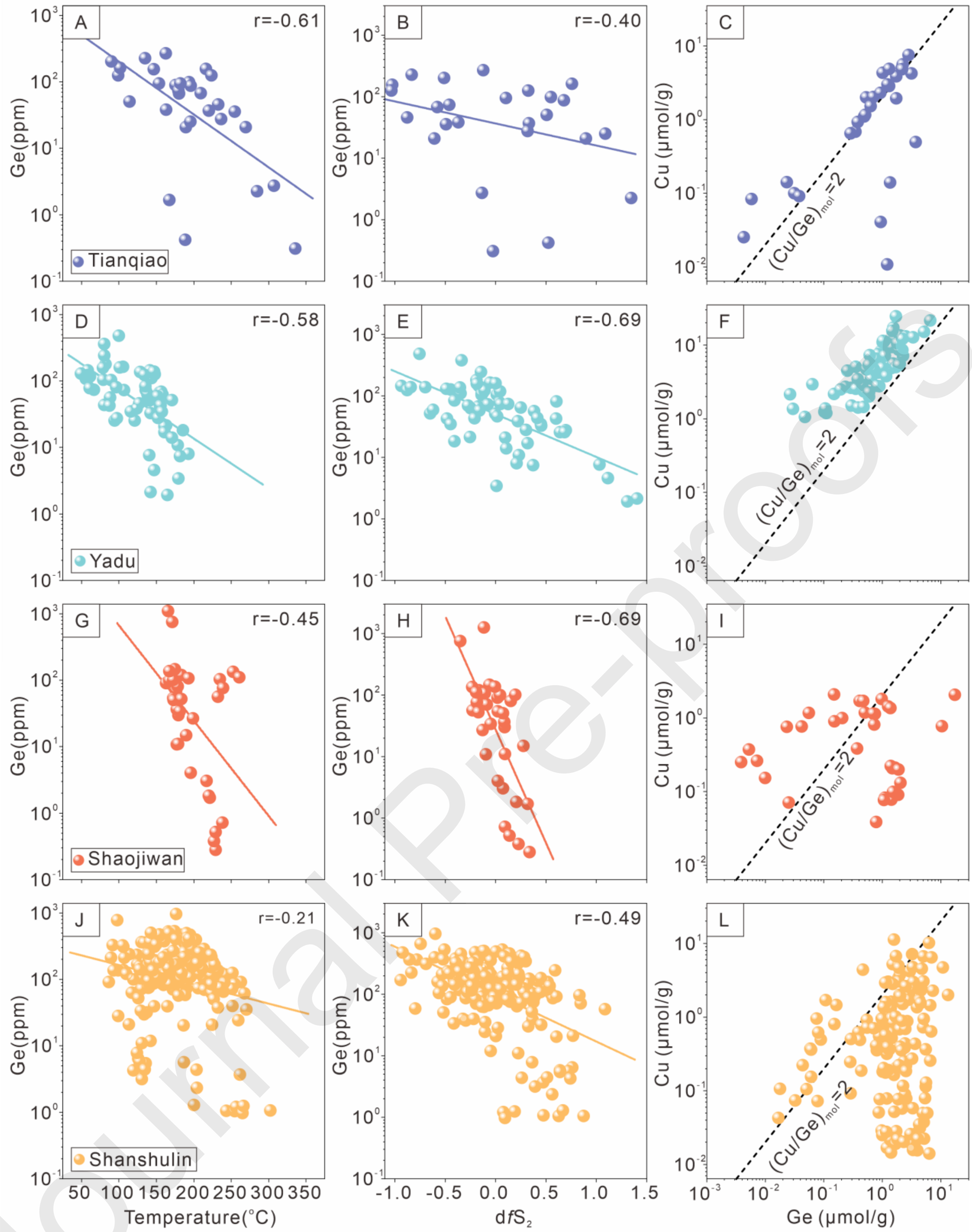

H+L+V liquid
1×10⁵ Pa
5
10
20
40
60
80
100
120
20%
15%
10%
liquid
pure H_2O
Critical curve
vapor
H+L+V vapor
ρ/(g·cm⁻³)
Th/°C
1.0
0.8
0.6
0.4
0.2
100
200
300
400
Tianqiao
Yadu
Shaojiwan
Shanshulin

Figure 14



A
r=-0.61
Tianqiao
B
r=-0.40
C
D
r=-0.58
Yadu
E
r=-0.69
F
G
r=-0.45
Shaojiwan
H
r=-0.69
I
J
r=-0.21
Shanshulin
K
r=-0.49
L
Ge(ppm)
Cu (μmol/g)
(Cu/Ge)mol=2
Temperature(°C)
dfS2
Ge (μmol/g)

Figure 15

Table1 Summary of LA-ICP-MS data of sphalerite from the Shanshulin Pb-Zn deposit (data in ppm)

Subtype		**Mn**	**Fe**	**Co**	**Ni**	**Cu**	**Ga**	**Ge**	**As**	**Ag**	**Cd**	**In**	**Sn**	**Sb**	**Te**	**Tl**	**Pb**
Sp1a（n=88）	Average	113	23461	0.20	2.15	57.4	9.25	169	3.60	99.7	765	0.08	1.99	6.53	1.94	1.06	46.1
	Median	81.5	22375	0.09	0.90	7.25	1.04	139	1.36	47.4	682	0.01	0.57	2.44	1.95	0.55	17.8
	Min	3.56	1421	b.d.l	b.d.l	b.d.l	b.d.l	b.d.l	b.d.l	1.91	203	b.d.l	b.d.l	b.d.l	b.d.l	b.d.l	b.d.l
	Max	655	49086	0.97	7.42	648	280	533	11.2	651	2069	1.31	11.6	73.6	4.02	4.84	615
Sp1b（n=73）	Average	69.9	12048	0.09	0.43	95.6	20.8	210	1.35	146	878	0.25	10.6	4.92	1.86	2.70	66.5
	Median	21.7	10051	0.07	0.47	46.3	3.29	163	1.03	92.6	715	0.02	0.91	1.81	1.85	0.61	13.5
	Min	2.27	1808	b.d.l	b.d.l	b.d.l	b.d.l	1.29	b.d.l	1.42	179	b.d.l	b.d.l	b.d.l	b.d.l	b.d.l	b.d.l

	Max	762	51566	0.11	0.56	718	424	962	2.81	636	2053	2.98	122	32.5	3.02	55.4	1181
Sp2（n=38）	Average	21.5	5310	0.06	0.93	81.3	13.1	126	1.42	77.3	1087	0.19	14.6	2.90	1.32	0.46	7.88
	Median	7.29	4000	0.06	0.93	44.3	8.43	98.3	1.42	31.1	987	0.09	3.01	2.48	1.27	0.28	3.34
	Min	1.94	1655	b.d.l	b.d.l	b.d.l	b.d.l	4.29	b.d.l	0.92	273	b.d.l	b.d.l	b.d.l	b.d.l	b.d.l	b.d.l
	Max	355	37051	0.08	1.19	412	49.2	505	1.42	419	2115	0.81	54.7	11.1	2.01	1.15	50.3

Footnote: The abbreviation of “b.d.l.” stands for the data below the limit of detection.

Table1